DESIGN PROJECTS

Scenarios and Rules[1]

Mechanical Engineering

National University of Singapore

Prof. Cecilia Laschi, Dr. Hongying Zhang

1 The design project is developed with reference to the RoboSoft Competition 2023 guidelines and innovations.

DESIGN PROJECT OVERVIEW

The primary goal of the design project is to generate innovative ideas in soft robotics by applying the knowledge acquired from our courses. The project emphasizes the unique aspects of soft robots, such as resilience, body compliance, delicate contact, and deformability.

Team Composition: We strongly encourage randomly formed teams to ensure a diverse mix of students in terms of academic background and other criteria. It is also recommended to appoint a Team Leader (TL) to act as the main point of contact for each team.

Design Topics: The design project centers on practical applications of soft robots, specifically targeting tree challenges: “food handling for tray preparation”, “in-pipe locomotion”, and “controlling of a continuum manipulator”. These challenges are designed to reflect and utilize the advancements in soft robotics over the last decade. To be more specific on the controlling project, there are five EMIO robot (https://compliance-robotics.com/compliance-lab/) available at the control lab (E2-01-02).

Submissions: A physical prototype of the design is not mandatory for all teams; however, CAD models and simulations are essential to demonstrate the functionality and effectiveness of the proposed designs in addressing the chosen challenges. Each team is required to present their design during their pre-assigned timeslots. This presentation is an opportunity to showcase the innovative work you have accomplished, explain your design process, and highlight the unique features of your creation. Alongside your presentation, please submit your presentation slides to the CANVAS.

DESIGN PROJECTS

The Design Project is composed of two scenarios and an EMIO continuum robot described in detail in the following, and each group is required to **choose one project** and develop your soft robot accordingly.

I. TERRESTRIAL RACE

The robot is deployed inside a pipeline which requires inspection. The pipes are connected to create a maze, the robot has to demonstrate the capability to autonomously navigate from start to exit in **simulation**. Five different joints are proposed: straight, curved, restrictions, junctions, flexibles, and polluted. The straight pipes have small apertures that can be used to recover or move the robot, from task to task:

- Task 1: The robot starts on a straight pipe;
- Task 2: A 45-degree turn;
- Task 3: A 90-degree turn;
- Task 4: A U-turn;
- Task 5: An obstructed sector: the diameter of the pipe is reduced by 50%;
- Task 6: A 90-degree turn upward, followed by another 90-degree;
- Task 7: A two junction: here the robot has to go straight initially, then turn to complete one of the branches proposed. It is not required to follow a specific order.
- Task 8: This part of the pipeline will have 5 cm silicone walls, so they will not provide a solid wall to push against;
- Task 9: This task comprises the movements on dirty sectors, to simulate realistic pipes, and it will conclude with the selection and exit from the last tract.

The whole track is available as .stp and .igs files to be downloaded from the CANVAS system.

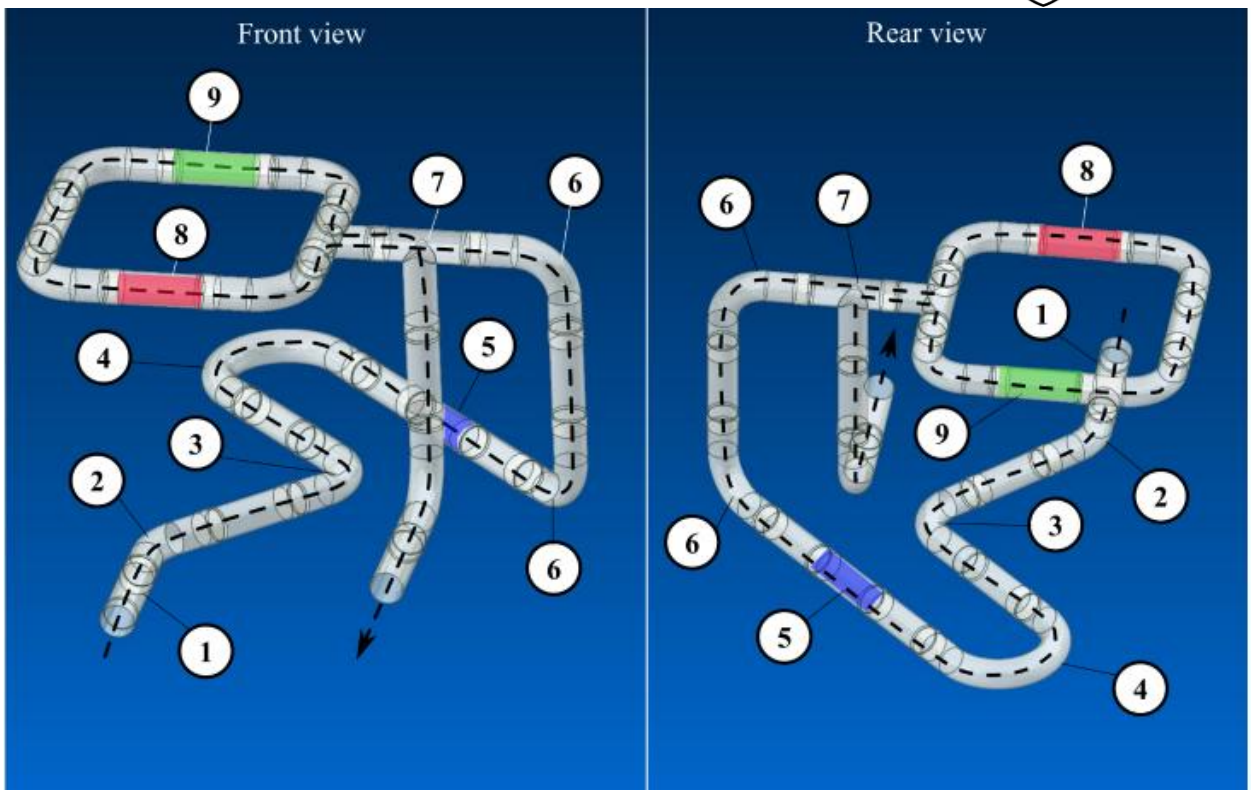

Figure 1 The pipeline proposed for the locomotion scenario. The robot has to autonomously navigate from the entrance to the exit (indicated by an arrow). The tubes are approximately 10 cm in diameter.

II. MANIPULATION CHALLENGE: Food Tray Assembly

The manipulation tasks focus on testing advances in **soft gripper designs** aiming to manipulating objects varying in size, texture, material and weight. For this, the tasks would revolve around putting together a tray of goods, a scenario applicable in healthcare, air travel, food and beverage, hospitality, and other service industries.

The soft gripper must have the capability to handle all food types in the **simulation**, that includes:

1. Sausages (cooked)
2. Meatballs (cooked)
3. Broccoli (florets, raw)
4. Carrots (sliced, 2-5 mm thick, cooked)
5. Green Beans or French String Beans (cooked)

6. Spaghetti Noodles (cooked)
7. Cookies
8. Fried eggs (sunny side up)
9. Orange Juice

Suppose a robotic arm can fulfill the picking task is in no specific order from one table and place them into the particular containers on a tray. There are 2 tables in the simulation scenario: one to hold the goods (Source Table) and another to keep the food trays to be assembled. Figure 2 shows the overall configuration and layout. The numbered shapes in both tables refer to indicative locations and not to actual shapes or measurements. The food and drink will be stored in good containers and jugs either on the table or on the shelf on the table at the source table. Each container will hold multiple units of the same food item. At the assembly table, two identical empty trays, with different sets of color plates/bowls/cups will be set up on the assembly table. Each transparent plastic container and side dish container will contain 1 food type. The cup is to be used for juice only.

Figure 2 Diagram of the manipulation set-up and layout of assembly and source tables.

III. CONTINUUM SOFT ROBOT CONTROL: EMIO robot

The EMIO robot is a hands-on platform for learning continuum and deformable robotics. It integrates with the SOFA simulation framework to demonstrate direct and inverse deformation modeling, simulation, and task-level control. Demos and details are available at https://compliance-robotics.com/compliance-lab/.

You may propose a specific task and control the platform to execute it. Availability is limited to five test sets on a first-come, first-served basis; please contact the course coordinators to check availability.

Figure 3 EIMO soft continuum robot

Skills to be shown in the project/Evaluation Criteria

Skills to be shown are directly related to scenarios and tasks: teams should demonstrate the minimum capabilities of their robots allowing them to be competitive. A complete list is presented here, grouped by scenario:

1. Terrestrial race
    a. Simulation of locomotion inside a straight pipeline

   b. Simulation of locomotion inside a curved pipeline (minimum 45 degrees, arbitrary curvature)

2. Manipulation
   a. Design of a soft gripper (by grasping, curling around, etc…)
   b. Compliance with the gripper
3. Control
   a. Forward and inverse kinematics in SOFA simulation framework
   b. Control of a continuum soft robot

The evaluation criteria will be on a do-it basis, thus **a simulation** demonstrating the ability to perform one of the skills above grants the grade. Fabrication will not be required at this stage, but please bear in mind that robots should be manufacturable using the techniques covered by the lecturers.

Reference

[1] https://robosoft2023.org/2023-competitions/

[2] https://compliance-robotics.com/compliance-lab/